



**Hochschule für Technik
und Wirtschaft Berlin**

University of Applied Sciences

Computer Networks

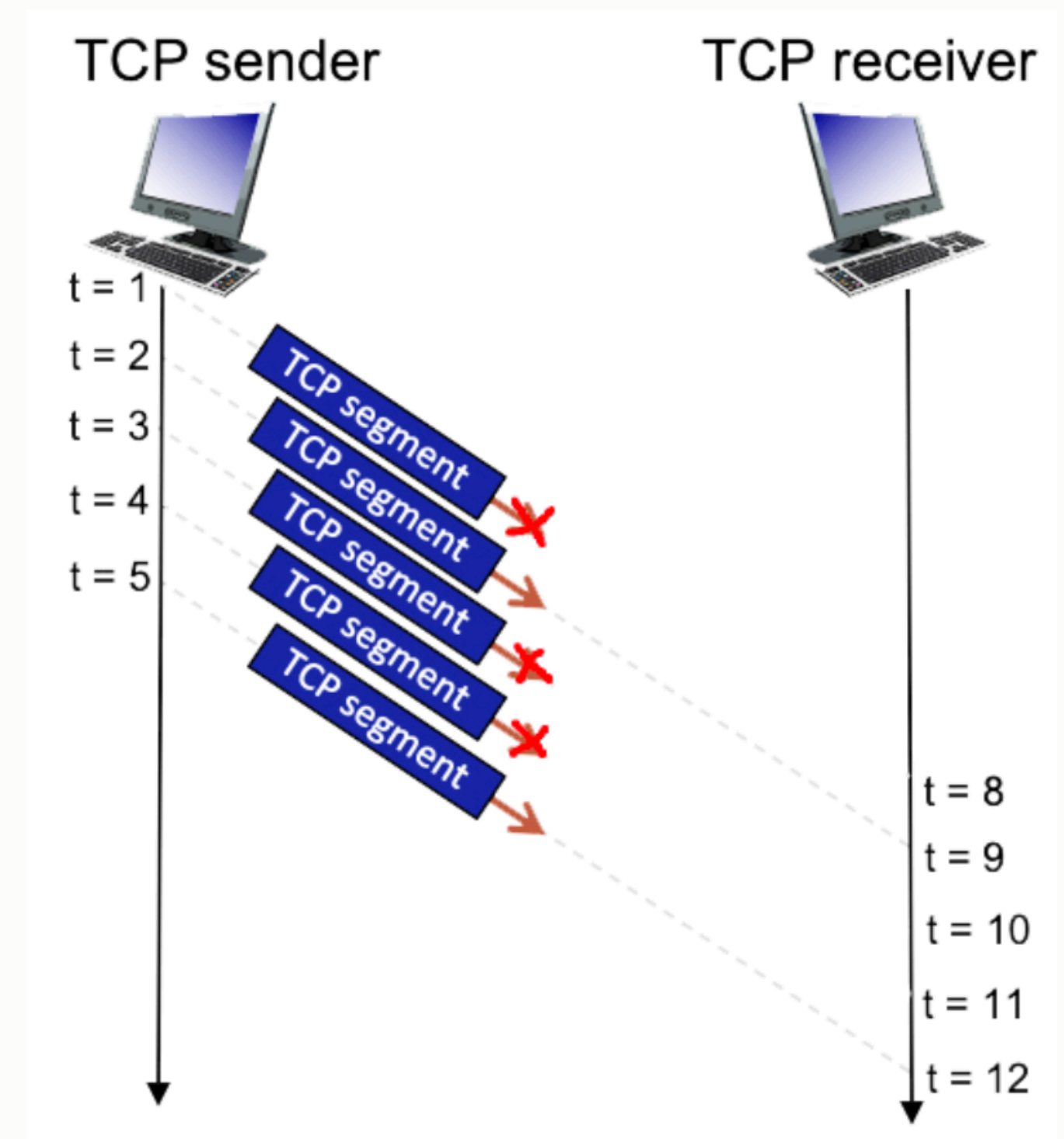
Malik Algazaeery

Lab-5

TCP sequence and ACK numbers with segment loss

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the sender->receiver segments may be lost. The TCP sender sends an initial window of 5 segments. Suppose the initial value of the sender->receiver sequence number is 78 and the first 5 segments each contain 720 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t=8$. As shown in the figure below, 3 of the 5 segment(s) are lost between the segment and receiver.

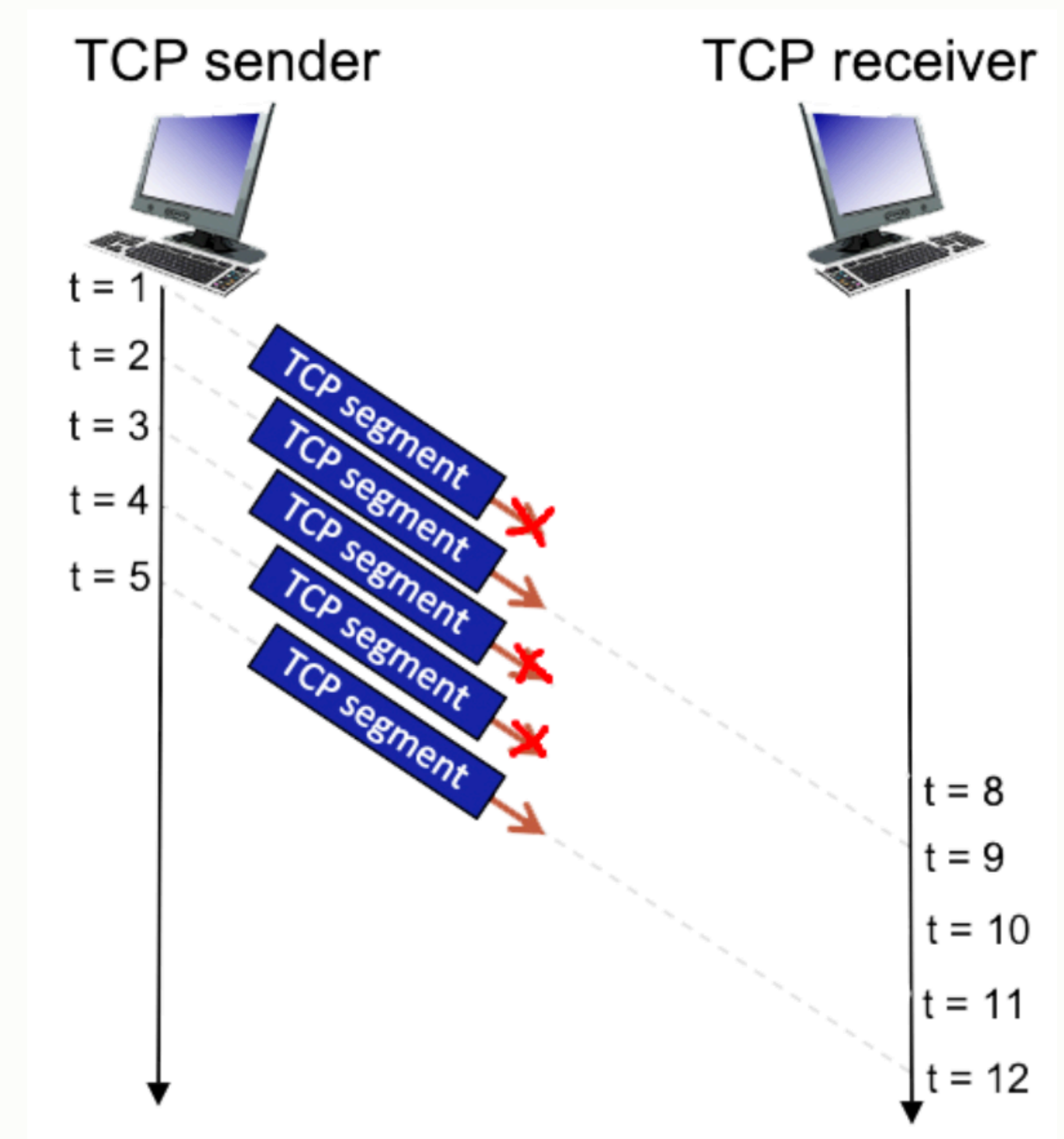
- Give the sequence numbers associated with each of the 5 segments sent by the sender. Format your answer as: a,b,c,...
- Give the ACK numbers the receiver sends in response to each of the segments. If a segment never arrives use 'x' to denote it, and format your answer as: a,b,c,...



TCP sequence and ACK numbers with segment loss

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the sender->receiver segments may be lost. The TCP sender sends an initial window of 5 segments. Suppose the initial value of the sender->receiver sequence number is 78 and the first 5 segments each contain 720 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t=8$. As shown in the figure below, 3 of the 5 segment(s) are lost between the segment and receiver.

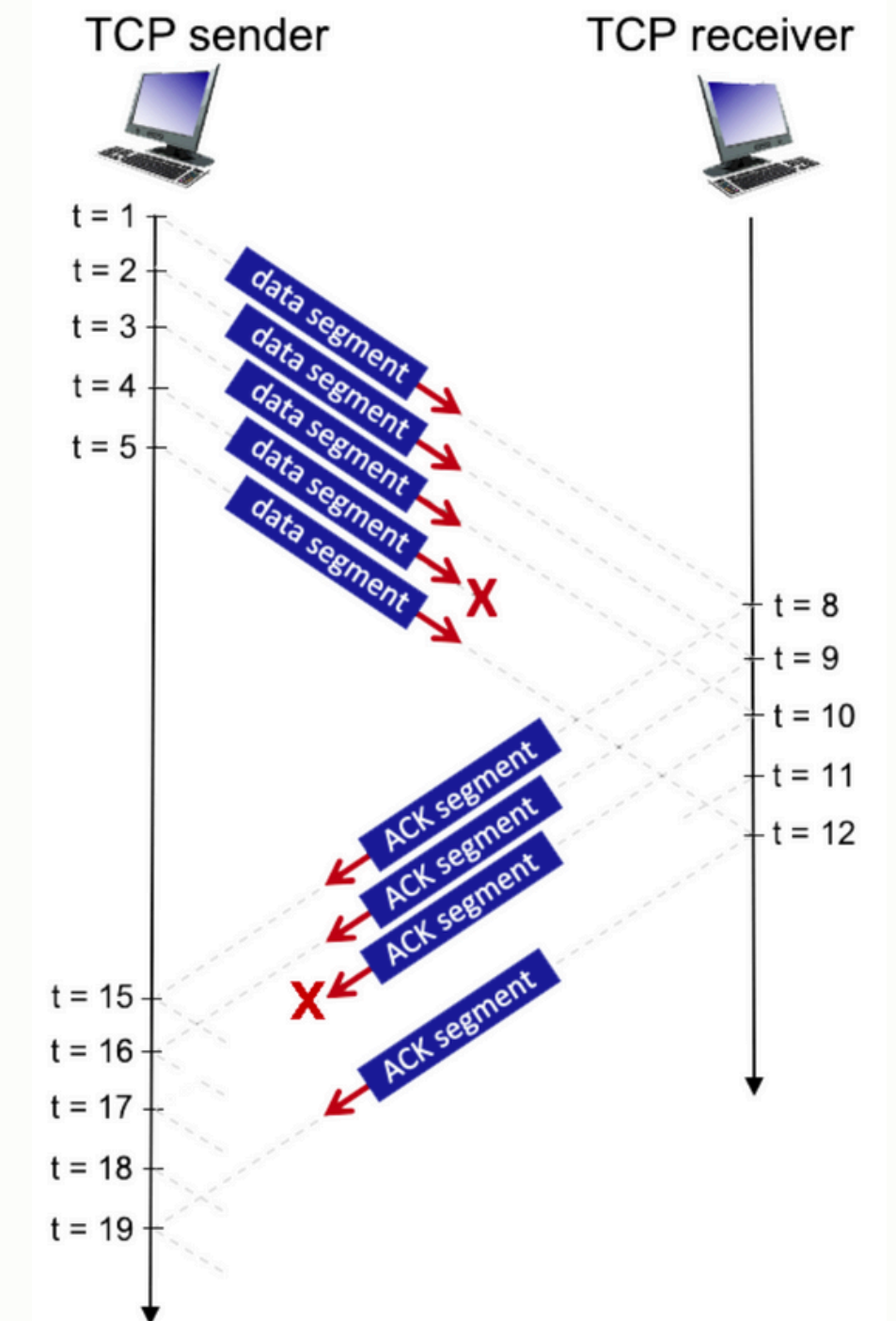
- Give the sequence numbers associated with each of the 5 segments sent by the sender. Format your answer as: a,b,c,...
 - The sender's sequence numbers are: 78,798,1518,2238,2958
- Give the ACK numbers the receiver sends in response to each of the segments. If a segment never arrives use 'x' to denote it, and format your answer as: a,b,c,...
 - The receiver's ACKs are: x,78,x,x,78



TCP retransmissions

(reliable data transmission with ACK loss)

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at $t = 1, 2, 3, 4,$ and 5 , respectively. Suppose the initial value of the sequence number is 102 and every segment sent to the receiver each contains 377 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t = 8$, and an ACK for this segment arrives at $t = 15$. As shown in the figure, 1 of the 5 segments is lost between the sender and the receiver, but one of the ACKs is lost. Assume there are no timeouts and any out of order segments received are thrown out.

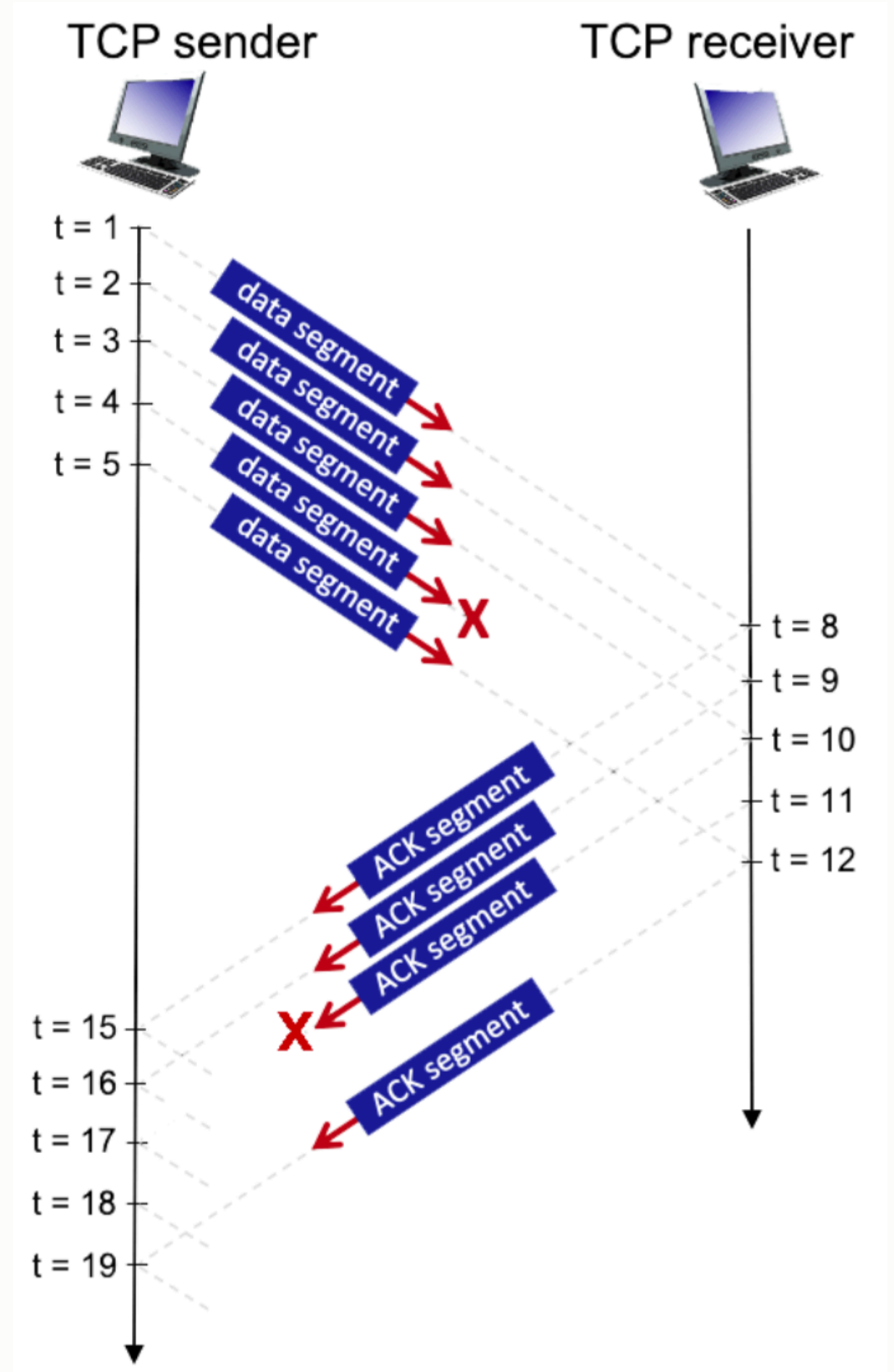


1. What is the sequence number of the segment sent at $t=1$?
2. What is the sequence number of the segment sent at $t=2$?
3. What is the sequence number of the segment sent at $t=3$?
4. What is the sequence number of the segment sent at $t=4$?
5. What is the sequence number of the segment sent at $t=5$?
6. What is the value of the ACK sent at $t=8$? (If segment lost, write 'x')
7. What is the value of the ACK sent at $t=9$? (If segment lost, write 'x')
8. What is the value of the ACK sent at $t=10$? (If segment lost, write 'x')
9. What is the value of the ACK sent at $t=11$? (If segment lost, write 'x')
10. What is the value of the ACK sent at $t=12$? (If segment lost, write 'x')
11. What is the sequence number of the segment sent at $t = 15$? (If ACK never arrives, write 'x')
12. What is the sequence number of the segment sent at $t = 16$? (If ACK never arrives, write 'x')
13. What is the sequence number of the segment sent at $t = 17$? (If ACK never arrives, write 'x')
14. What is the sequence number of the segment sent at $t = 18$? (If ACK never arrives, write 'x')
15. What is the sequence number of the segment sent at $t = 19$? (If ACK never arrives, write 'x')

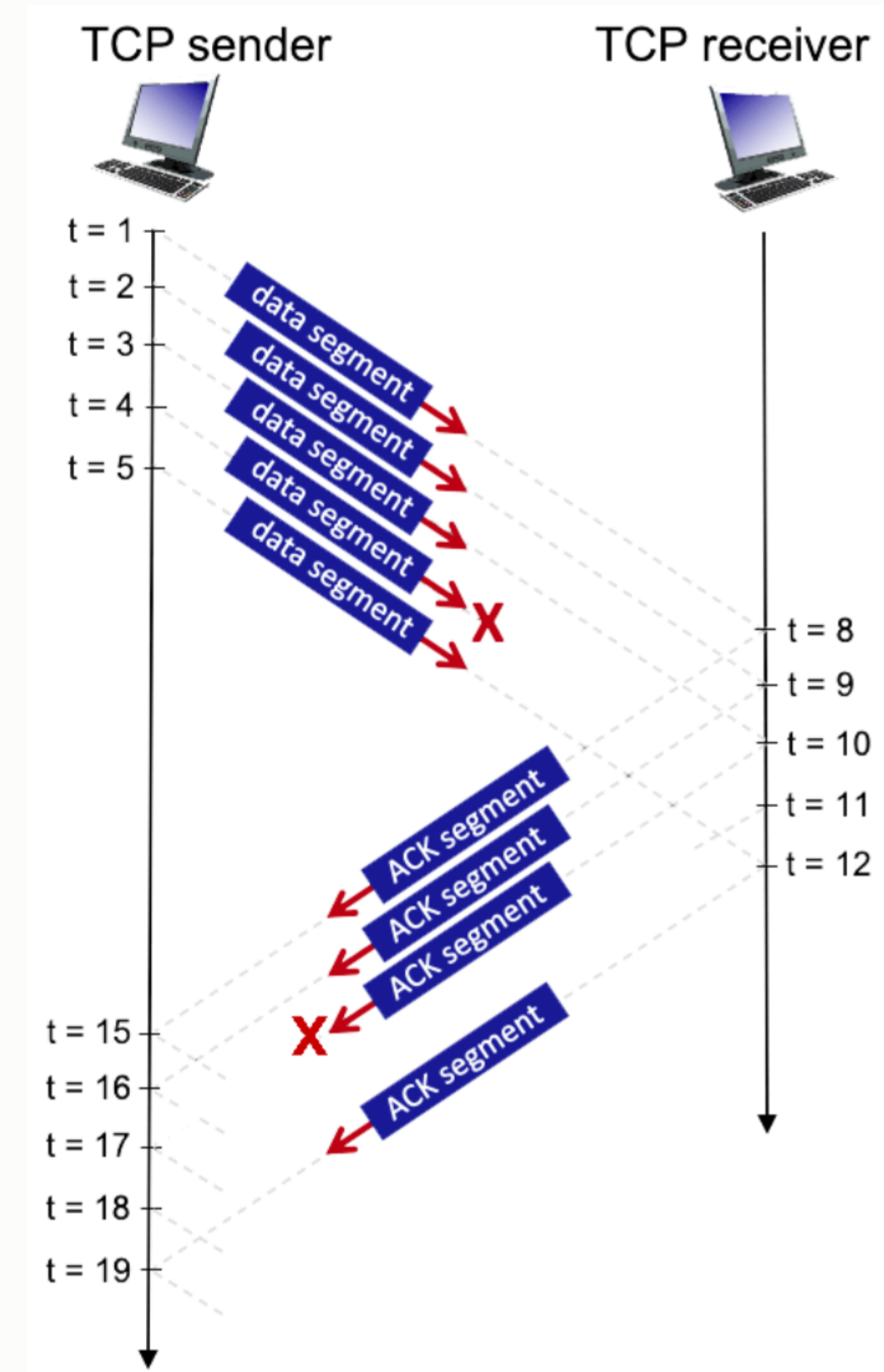
TCP retransmissions (reliable data transmission with ACK loss)

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at $t = 1, 2, 3, 4,$ and 5 , respectively. Suppose the initial value of the sequence number is 102 and every segment sent to the receiver each contains 377 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t = 8$, and an ACK for this segment arrives at $t = 15$. As shown in the figure, 1 of the 5 segments is lost between the sender and the receiver, but one of the ACKs is lost. Assume there are no timeouts and any out of order segments received are thrown out.

1. What is the sequence number of the segment sent at $t=1$?
 - The sequence number of the first segment is the starting sequence number, which is 102.
2. What is the sequence number of the segment sent at $t=2$?
 - The sequence number of the second segment is $= 102 + 377 = 479$.
3. What is the sequence number of the segment sent at $t=3$?
 - The sequence number of the third segment is $= 479 + 377 = 856$.
4. What is the sequence number of the segment sent at $t=4$?
 - The sequence number of the fourth segment is $= 856 + 377 = 1233$.
5. What is the sequence number of the segment sent at $t=5$?
 - The sequence number of the fifth segment is $= 1233 + 377 = 1610$.

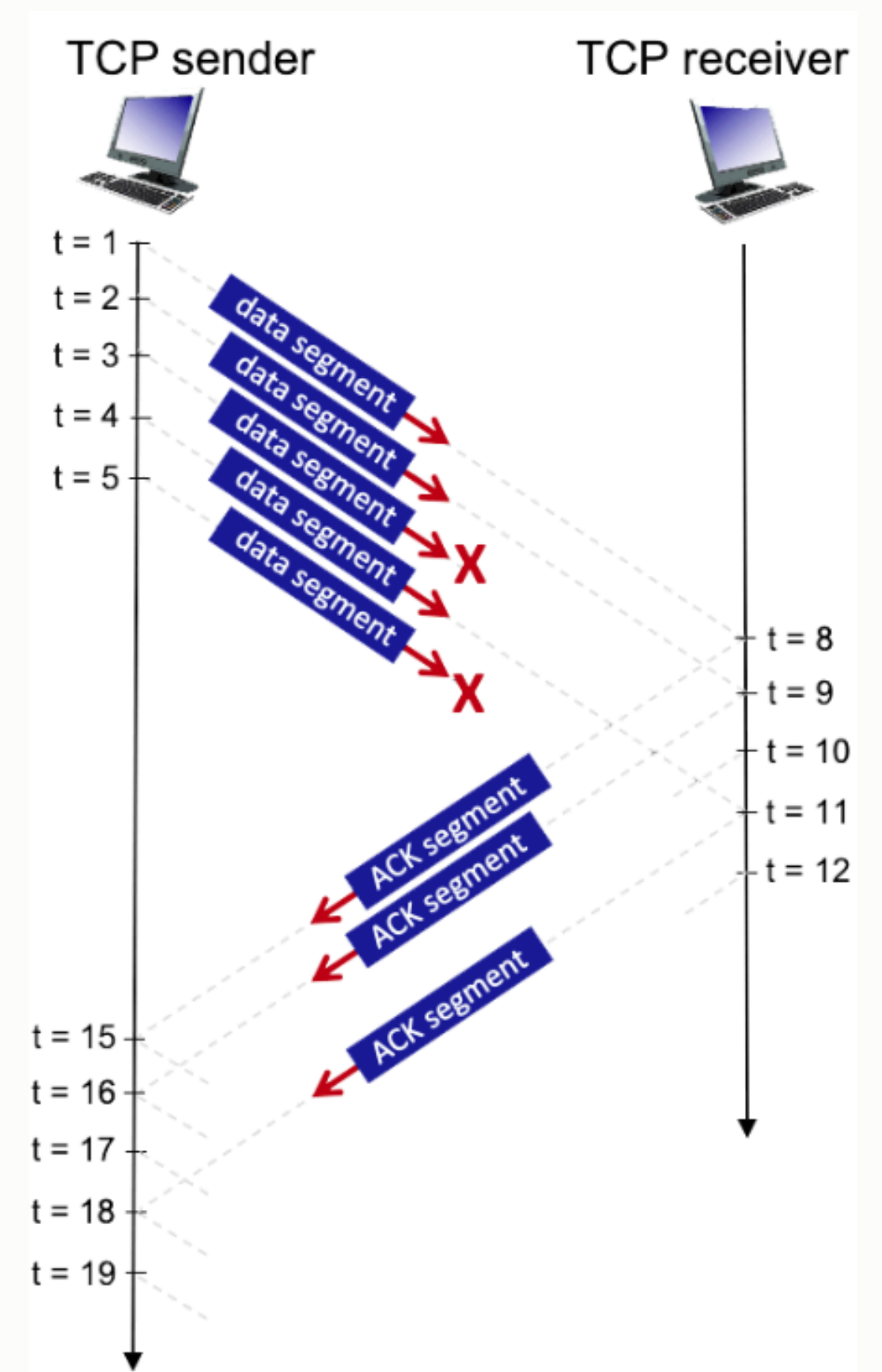


6. What is the value of the ACK sent at t=8? (If segment lost, write 'x')
- The ACK value is the sequence number of the next expected segment, which is 479.
7. What is the value of the ACK sent at t=9? (If segment lost, write 'x')
- The ACK value is the sequence number of the next expected segment, which is 856.
8. What is the value of the ACK sent at t=10? (If segment lost, write 'x')
- The ACK value is the sequence number of the next expected segment, which is 1233.
9. What is the value of the ACK sent at t=11? (If segment lost, write 'x')
- Since the segment was lost, the ACK is never sent, so the answer is x.
10. What is the value of the ACK sent at t=12? (If segment lost, write 'x')
- The ACK value is the sequence number of the next expected segment, which is 1233.
11. What is the sequence number of the segment sent at t = 15? (If ACK never arrives, write 'x')
- The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 1987.
12. What is the sequence number of the segment sent at t = 16? (If ACK never arrives, write 'x')
- The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 2364.
13. What is the sequence number of the segment sent at t = 17? (If ACK never arrives, write 'x')
- Since there was nothing to send (as we're waiting for ACKs), the answer is x.
14. What is the sequence number of the segment sent at t = 18? (If ACK never arrives, write 'x')
- Since there was nothing to send (as we're waiting for ACKs), the answer is x.
15. What is the sequence number of the segment sent at t = 19? (If ACK never arrives, write 'x')
- The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 2741.



Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at $t = 1, 2, 3, 4$, and 5 , respectively. Suppose the initial value of the sequence number is 76 and every segment sent to the receiver each contains 631 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at $t = 8$, and an ACK for this segment arrives at $t = 15$. As shown in the figure, 2 of the 5 segments is lost between the sender and the receiver, but none of the ACKs are lost. Assume there are no timeouts and any out of order segments received are thrown out.

1. What is the sequence number of the segment sent at $t=1$?
2. What is the sequence number of the segment sent at $t=2$?
3. What is the sequence number of the segment sent at $t=3$?
4. What is the sequence number of the segment sent at $t=4$?
5. What is the sequence number of the segment sent at $t=5$?
6. What is the value of the ACK sent at $t=8$? (If segment lost, write 'x')
7. What is the value of the ACK sent at $t=9$? (If segment lost, write 'x')
8. What is the value of the ACK sent at $t=10$? (If segment lost, write 'x')
9. What is the value of the ACK sent at $t=11$? (If segment lost, write 'x')
10. What is the value of the ACK sent at $t=12$? (If segment lost, write 'x')
11. What is the sequence number of the segment sent at $t = 15$? (If ACK never arrives, write 'x')
12. What is the sequence number of the segment sent at $t = 16$? (If ACK never arrives, write 'x')
13. What is the sequence number of the segment sent at $t = 17$? (If ACK never arrives, write 'x')
14. What is the sequence number of the segment sent at $t = 18$? (If ACK never arrives, write 'x')
15. What is the sequence number of the segment sent at $t = 19$? (If ACK never arrives, write 'x')

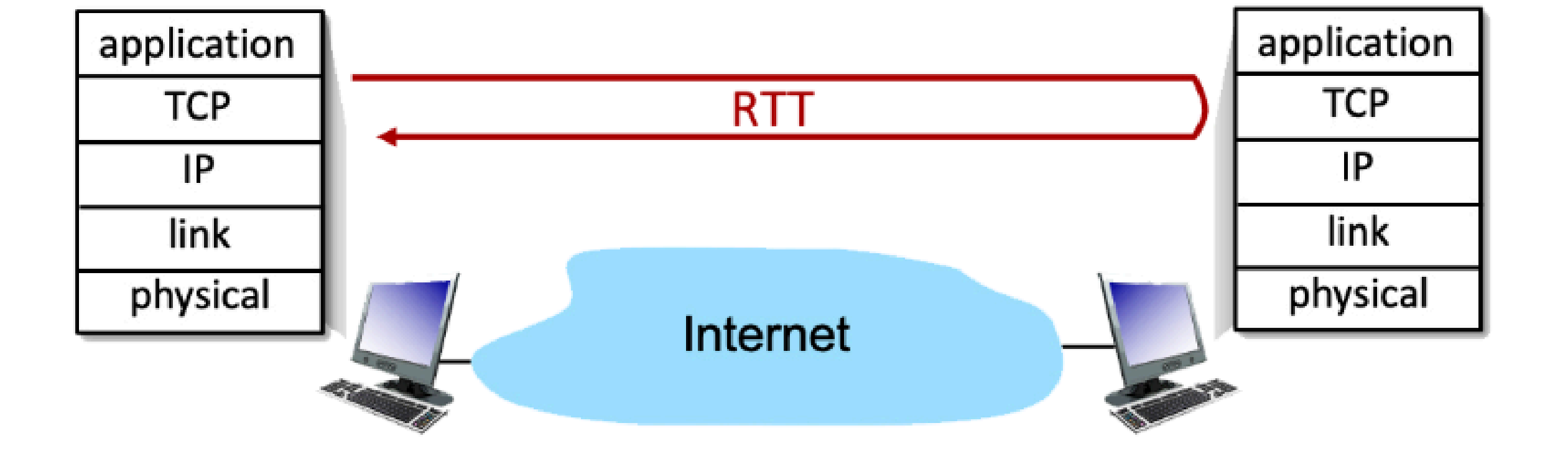


SOLUTION

1. The sequence number of the first segment is the starting sequence number, which is 76.
2. The sequence number of the second segment is $= 76 + 631 = 707$.
3. The sequence number of the third segment is $= 707 + 631 = 1338$.
4. The sequence number of the fourth segment is $= 1338 + 631 = 1969$.
5. The sequence number of the fifth segment is $= 1969 + 631 = 2600$.
6. The ACK value is the sequence number of the next expected segment, which is 707.
7. The ACK value is the sequence number of the next expected segment, which is 1338.
8. Since the segment was lost, the ACK is never sent, so the answer is x.
9. The ACK value is the sequence number of the next expected segment, which is 1338.
10. Since the segment was lost, the ACK is never sent, so the answer is x.
11. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 3231.
12. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 3862.
13. Since there was nothing to send (as we're waiting for ACKs), the answer is x.
14. Since there was nothing to send (as we're waiting for ACKs), the answer is x.
15. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

COMPUTING TCP'S RTT AND TIMEOUT VALUES

Suppose that TCP's current estimated values for the round trip time (*estimatedRTT*) and deviation in the RTT (*DevRTT*) are 320 msec and 25 msec, respectively (see Section 3.5.3 for a discussion of these variables). Suppose that the next three measured values of the RTT are 230 msec, 280 msec, and 270 msec respectively.



Compute TCP's new value of *DevRTT*, *estimatedRTT*, and the TCP timeout value after each of these three measured RTT values is obtained. Use the values of $\alpha = 0.125$, and $\beta = 0.25$. Round your answers to two decimal places after leading zeros.

SOLUTION

DevRTT is calculated with the following equation: $(1-\beta) \cdot \text{DevRTT} + \beta \cdot |\text{estimatedRTT} - \text{sampleRTT}|$

estimatedRTT is calculated with the following equation: $(1-\alpha) \cdot \text{estimatedRTT} + \alpha \cdot \text{sampleRTT}$

TCP timeout is calculated with the following equation: $\text{estimatedRTT} + (4 \cdot \text{DevRTT})$

1. The estimatedRTT for RTT1 is 308.75

2. The DevRTT for RTT1 is 41.25

3. The timeout for RTT1 is 473.75

4. The estimatedRTT for RTT2 is 305.16

5. The DevRTT for RTT2 is 38.13

6. The timeout for RTT2 is 457.66

7. The estimatedRTT for RTT3 is 300.76

8. The DevRTT for RTT3 is 37.38

9. The timeout for RTT3 is 450.29